EFFECTIVITY OF DESCENT DATA DESCENDS ALONG A FAITHFULLY FLAT EXTENSION OF THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Criterion for recognizing effectivity of a descent datum. Same chapter, Lemma lemma-recognize-effective, invoked at proof lines 850. Supplies the conclusion from the base changed datum being effective.
- Base change of a descent datum along a ring map and the resulting identification. Standard descent algebra, used at proof lines 840-849. Supplies the objects N' and phi' and the comparison of the two canonical maps.

Source reference. Original lemma 039W: sources/stacks-descent/pinned-original-source.tex, source lines 807–807. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let $R \to A$ be a faithfully flat ring map, and let $R \to R'$ be faithfully flat. Set $A' = R' \otimes_R A$. If all descent data for $R' \to A'$ are effective, then so are all descent data for $R \to A$.*

Proof. Let (N, φ) be a descent datum for $R \to A$. Set $N' = R' \otimes_R N = A' \otimes_A N$, and denote $\varphi' = \mathrm{id}_{R'} \otimes \varphi$ the base change of the descent datum φ. Then (N', φ') is a descent datum for $R' \to A'$ and $H^0(s(N'_\bullet)) = R' \otimes_R H^0(s(N_\bullet))$. Moreover, the map $A' \otimes_{R'} H^0(s(N'_\bullet)) \to N'$ is identified with the base change of the A-module map $A \otimes_R H^0(s(N)) \to N$ via the faithfully flat map $A \to A'$. Hence we conclude by Lemma 039W. $\square$

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